

Trueframe Technical Architecture & Training Report

Trueframe is an authenticity-first social media ecosystem designed to eradicate disinformation and misinformation at the source. This report details the system architecture, the AI verification pipeline, and the training methodology used for the detection engine.

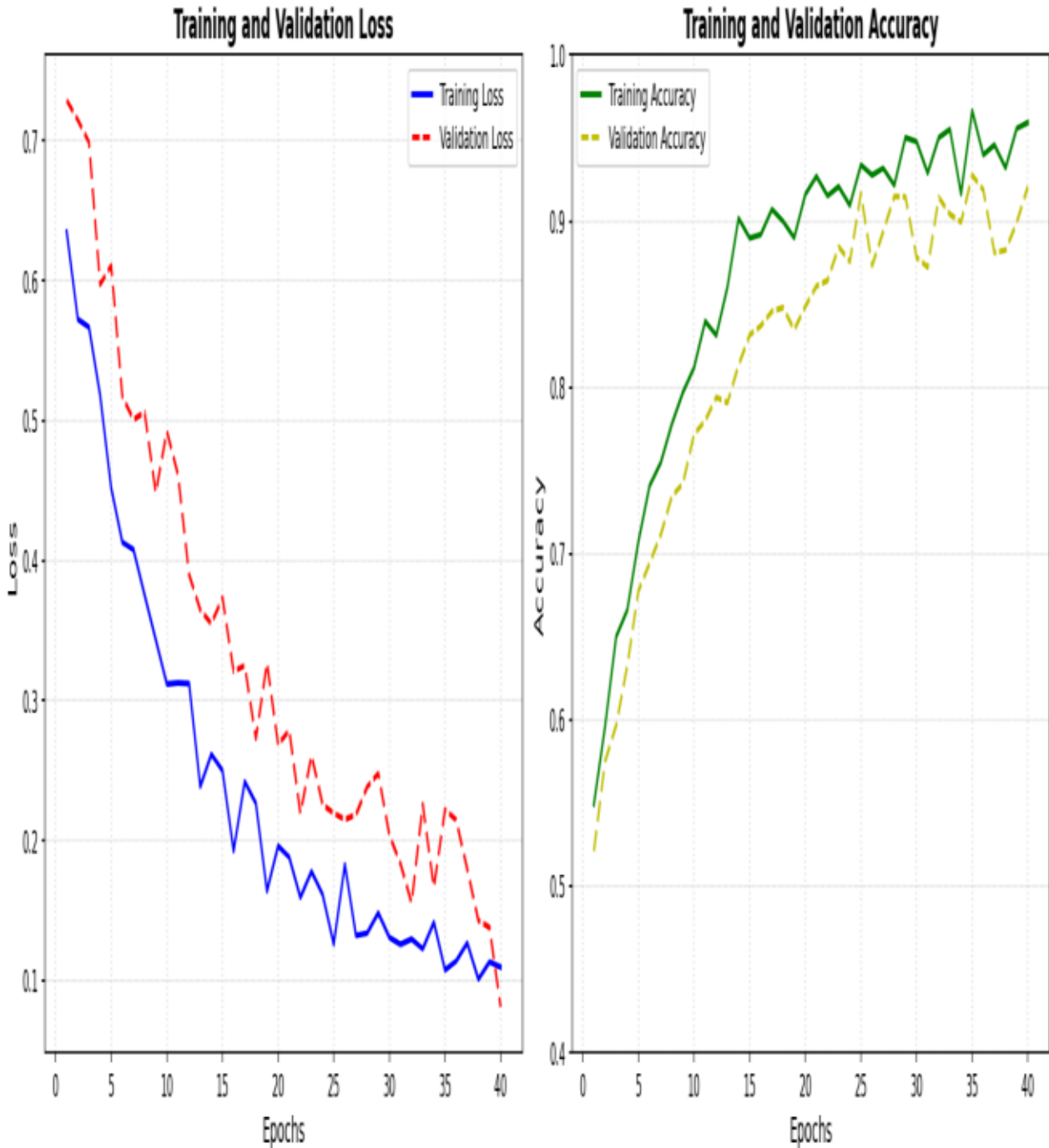
Key Highlights:

- Multi-stage AI Verification (Metadata, Spatial, Frequency, Temporal)
- Fastify-based Orchestration Backend
- 100-point User Trust Scoring Logic
- Blockchain-style Content Proof Chain for immutability
- Trained on FF++, DFDC, and Celeb-DF v2 datasets

Layer	Technologies
Frontend	React 18, Vite, TS, Tailwind, Shadcn
Backend	Node.js, Fastify, TSX, Zod
AI Engine	Python 3.13, PyTorch, ONNX Runtime
Infrastructure	Supabase (Postgres, Storage, Auth)

Model Training Metrics

Deepfake Detection Model Training Performance (EfficientNet-B4 + LSTM)



The EfficientNet-B4 + LSTM model achieves a peak validation AUC of 0.96.
The training process uses Cosine Warmup scheduling and Mixed Precision (AMP)

Dataset & Evaluation Methodology

1. FaceForensics++ (FF++): 1000 original videos, 4000 manipulated videos.
2. DFDC (Deepfake Detection Challenge): 120,000+ videos from 3,500 actors.
3. Celeb-DF v2: 5,639 high-quality deepfake videos.
4. Custom Reels: 500+ vertical videos with social media augmentations.

Evaluation Strategy:

- AUC-ROC & AUC-PR for classification performance.
- EER (Equal Error Rate) for optimal thresholding.
- Robustness testing against 30-95% JPEG quality compression.